

Math 122: Exam 1 Solutions

Problem 1 Find the exact arc length of the curve $y = \frac{2}{3}x^{3/2} + 2026$ over the interval from $x = 0$ to $x = 1$ [cite: 1]. * The formula for arc length is

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

* Calculate the derivative of the given function:

$$\frac{dy}{dx} = \frac{2}{3} \cdot \frac{3}{2} x^{1/2} = \sqrt{x}$$

* Square the derivative:

$$\left(\frac{dy}{dx}\right)^2 = (\sqrt{x})^2 = x$$

* Set up the integral with the given bounds:

$$L = \int_0^1 \sqrt{1+x} dx$$

* Evaluate using the substitution $u = 1 + x$ and $du = dx$:

$$L = \left[\frac{2}{3}(1+x)^{3/2} \right]_0^1 = \frac{2}{3}(2^{3/2} - 1)$$

* Exact answer:

$$\frac{2}{3}(2\sqrt{2} - 1)$$

* **Decimal answer: 1.21895**

Problem 2 Find the limit: $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sin(2x)}$ [cite: 2]. * Substituting $x = 0$ yields the indeterminate form $\frac{0}{0}$. * Apply L'Hôpital's Rule by taking the derivative of the numerator and the denominator:

$$\lim_{x \rightarrow 0} \frac{2^x \ln(2)}{2 \cos(2x)}$$

* Evaluate the limit as $x \rightarrow 0$:

$$\frac{2^0 \ln(2)}{2 \cos(0)}$$

* Exact answer:

$$\frac{\ln(2)}{2}$$

* **Decimal answer: 0.34657**

Problem 3 Find the limit: $\lim_{x \rightarrow +\infty} x^2 2^{-3x}$ [cite: 2]. * Rewrite the expression as a fraction:

$$\lim_{x \rightarrow +\infty} \frac{x^2}{2^{3x}}$$

* This yields the indeterminate form $\frac{\infty}{\infty}$, so apply L'Hôpital's Rule:

$$\lim_{x \rightarrow +\infty} \frac{2x}{3 \cdot 2^{3x} \ln(2)}$$

* Apply L'Hôpital's Rule a second time for the remaining $\frac{\infty}{\infty}$ form:

$$\lim_{x \rightarrow +\infty} \frac{2}{9 \cdot 2^{3x} (\ln(2))^2}$$

* As $x \rightarrow \infty$, the denominator grows without bound while the numerator is constant. *

Decimal answer: 0

Problem 4 Find the limit: $\lim_{x \rightarrow 0^+} (1 + 3x)^{1/x}$ [cite: 3]. * This limit takes the indeterminate form 1^∞ . * Let $y = (1 + 3x)^{1/x}$. Taking the natural logarithm gives $\ln(y) = \frac{\ln(1+3x)}{x}$. * Evaluate the limit of $\ln(y)$ as $x \rightarrow 0^+$, which takes the form $\frac{0}{0}$, using L'Hôpital's Rule:

$$\lim_{x \rightarrow 0^+} \frac{\frac{3}{1+3x}}{1} = 3$$

* Since $\lim_{x \rightarrow 0^+} \ln(y) = 3$, exponentiate to find the limit of y . * Exact answer:

$$e^3$$

* **Decimal answer: 20.08554**

Problem 5 Find du for $y = \sin^{-1}\left(\frac{1}{x^3}\right)$ [cite: 3]. * Apply the chain rule using the derivative of inverse sine:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - (x^{-3})^2}} \cdot \frac{d}{dx}(x^{-3})$$

* Evaluate the inner derivative:

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - x^{-6}}} \cdot (-3x^{-4})$$

* **Answer:**

$$du = -\frac{3}{x^4\sqrt{1-x^{-6}}}dx$$

(No decimal evaluation applies).

Problem 6 Find the derivative of $y = \tan^{-1}(x) \cdot \sec^{-1}(x)$ [cite: 4]. * Apply the product rule:

$$\frac{dy}{dx} = \frac{d}{dx}(\tan^{-1}(x)) \cdot \sec^{-1}(x) + \tan^{-1}(x) \cdot \frac{d}{dx}(\sec^{-1}(x))$$

* **Answer:**

$$\frac{dy}{dx} = \frac{\sec^{-1}(x)}{1+x^2} + \frac{\tan^{-1}(x)}{|x|\sqrt{x^2-1}}$$

(No decimal evaluation applies).

Problem 7 Evaluate the integral: $\int \frac{dx}{1+16x^2}$ [cite: 4]. * Rewrite the denominator:

$$\int \frac{dx}{1+(4x)^2}$$

* Use substitution with $u = 4x$, making $dx = \frac{du}{4}$. * Substitute and integrate:

$$\frac{1}{4} \int \frac{du}{1+u^2} = \frac{1}{4} \tan^{-1}(u) + C$$

* **Answer:**

$$\frac{1}{4} \tan^{-1}(4x) + C$$

(No decimal evaluation applies).

Problem 8 Evaluate the integral: $\int \frac{x^2 dx}{\sqrt{1-x^6}}$ [cite: 5]. * Rewrite the integral:

$$\int \frac{x^2}{\sqrt{1-(x^3)^2}} dx$$

* Let $u = x^3$, giving $du = 3x^2 dx$, or $x^2 dx = \frac{du}{3}$. * Substitute and integrate:

$$\frac{1}{3} \int \frac{du}{\sqrt{1-u^2}} = \frac{1}{3} \sin^{-1}(u) + C$$

* **Answer:**

$$\frac{1}{3} \sin^{-1}(x^3) + C$$

(No decimal evaluation applies).

Problem 9 Evaluate the definite integral: $\int_0^1 \frac{(\sin^{-1} x)^4}{\sqrt{1-x^2}} dx$ [cite: 5]. * Let $u = \sin^{-1}(x)$, which makes $du = \frac{dx}{\sqrt{1-x^2}}$. * Determine new limits: When $x = 0$, $u = 0$. When $x = 1$, $u = \frac{\pi}{2}$. * Evaluate the new integral:

$$\int_0^{\pi/2} u^4 du = \left[\frac{u^5}{5} \right]_0^{\pi/2}$$

* Exact answer:

$$\frac{(\pi/2)^5}{5} = \frac{\pi^5}{160}$$

* **Decimal answer: 1.91262**

Problem 10 Evaluate the integral: $\int \frac{e^{2x}}{1+e^{4x}} dx$ [cite: 6]. * Rewrite to identify the substitution:

$$\int \frac{e^{2x}}{1+(e^{2x})^2} dx$$

* Let $u = e^{2x}$, yielding $du = 2e^{2x} dx$, or $e^{2x} dx = \frac{du}{2}$. * Substitute and integrate:

$$\frac{1}{2} \int \frac{du}{1+u^2} = \frac{1}{2} \tan^{-1}(u) + C$$

* **Answer:**

$$\frac{1}{2} \tan^{-1}(e^{2x}) + C$$

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